

Standard Form & Indices

Test 5 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $3^7 \times 3^2$ as a single power of 3.

[1 mark]

A2. Write $8^{11} \div 8^3$ as a single power of 8.

[1 mark]

A3. Write $(6^3)^3$ as a single power of 6.

[1 mark]

A4. Work out the value of $3^2 \times 3^2$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 23^0 .

[1 mark]

B2. Work out the value of 6^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-2} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 47,000 in standard form.

[2 marks]

C2. Write 2,800,000 in standard form.

[2 marks]

C3. Write 0.0074 in standard form.

[2 marks]

C4. Write 0.0003 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 7.2×10^5 as an ordinary number.

[1 mark]

D2. Write 4×10^{-3} as an ordinary number.

[1 mark]

D3. Work out $(5 \times 10^4) \times (8 \times 10^2)$. Give your answer in standard form.

[3 marks]

D4. Work out $(2 \times 10^7) \div (5 \times 10^3)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 3^5 .

[1 mark]

E2. $\sqrt{75}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $p^6 \times p^4$.

[1 mark]

F2. Simplify $q^{15} \div q^7$.

[1 mark]

F3. Simplify $(r^3)^5$.

[1 mark]

F4. Simplify $4h^2 \times 7h^6$.

[2 marks]

F5. Simplify $(5j^3)^2$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $81^{1/2}$.

[1 mark]

G2. Work out the value of $1000^{1/3}$.

[1 mark]

G3. Work out the value of $125^{2/3}$.

[2 marks]

G4. Work out the value of $8^{-2/3}$.

[2 marks]